3.1

ADULTS

BONE AND JOINT INFECTIONS

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Risk factors for consideration of empirical MRSA coverage[1]

1. Local prevalence
2. Patient-specific risk factors, such as known colonization status, prior positive cultures
3. Health care exposure

In a setting with low MRSA incidence, no known MRSA colonization or prior positive cultures, and minimal health care contact, it is reasonable to withhold empirical MRSA coverage.

Risk factors for consideration of empirical *Pseudomonas* spp. coverage[1]

Patients with chronic wounds who have:

1. been exposed to multiple prior courses of antibiotics
2. previously had cultures positive for *P. aeruginosa* within last 3 months
3. gangrenous wounds
4. had a recent surgical procedure (e.g., within 3 months, as with early PJI)
5. specific sites of infection particularly associated with pseudomonal infection (e.g., malignant otitis externa) within last 3 months

When to add rifampicin?[2]

(**Note:** Must be combined with another active antibacterial agent because rapid emergence of resistance in the setting of rifampicin monotherapy)

1. Rifampicin should be initiated several days after surgical debridement (after wound is dry); resistance is less likely to emerge when the burden of organisms has been reduced.
2. Continue rifampicin for the duration of definitive therapy (parenteral therapy or highly bioavailable oral antibiotic therapy).
3. Do not use rifampicin during suppressive therapy.
4. Do not use other rifamycins (e.g., rifabutin) routinely in the setting of rifampicin intolerance although in vitro and in vivo data are promising but clinical trials are warranted.
5. Suggest to monitor liver function test while on rifampicin and check for drug-drug interactions.

References:

1. Use of Novel Strategies to Develop Guidelines for Management of Pyogenic Osteomyelitis in Adults. JAMA Network Open, 5(5), e2211321. https://doi.org/10.1001/jamanetworkopen.2022.11321.
2. UpToDate: Nonvertebral osteomyelitis in adults: Treatment. Date accessed: 20 March 2024.

CHAPTER 3.1	BONE AND JOINT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Septic arthritis			
Acute monoarticular (In person who do not have any risk factors for STD) *S. aureus* *Streptococcus spp.*	Cloxacillin 2 g IV q4-6H	Cefazolin 2 g IV q6-8H **Penicillin Allergy:** Clindamycin 600 mg IV q8H	• Drainage, debridement, and washout of infected joints is important to limit further damage. • Empirical therapy wherever possible should be directed by the result of the Gram stain of the joint aspirate. • **Duration:** • Initial IV therapy: 2 weeks. • Oral therapy: 2-4 weeks. • Total duration: 4-6 weeks. • For **IV to oral conversion**, see notes below. • For patients with septic arthritis associated with an animal bite, the approach to antibiotic selection is discussed at **Chapter 3.10: Skin and Soft Tissue Infections.**
Acute monoarticular (In person who have risk factors for STD) *Gonorrhoea* *Streptococcus* *Staphylococcus* Gram-negative Bacilli	Ceftriaxone 2 g IV q24H for 1-2 weeks + Doxycycline 100 mg PO q12H for 7 days	Ceftriaxone 2 g IV q24H for 1-2 weeks + Azithromycin 1 g PO single dose	

CHAPTER 3.1	BONE AND JOINT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Osteomyelitis			
Acute osteomyelitis *S. aureus* (80%) *Streptococcus pyogenes* Gram-negative Bacilli (rare)	Cloxacillin 2 g IV q6H	Cefazolin 2 g IV q8H **Penicillin Allergy:** Clindamycin 600 mg IV q8H **OR** Vancomycin 15 mg/kg (max 1 g) IV q8-12H	• **Duration:** • **If the infected bone is not fully resected:** ○ Initial IV therapy for 2 weeks, then followed by oral therapy. ○ Total duration of minimum 6 weeks. • **A shorter duration of antibiotics can be considered if the infected bone is fully resected (e.g., amputation with a clear margin)[1]:** ○ No surrounding soft tissue infection: 5 days. ○ Evidence of soft tissue infection: 10-14 days. • Tailored to tissue culture and sensitivity as swab culture is not reliable. • Suggest to send bone biopsy at the clean margin intra-operatively, for histopathology and culture and sensitivity. • Surgical debridement for management of osteomyelitis consists of removal of necrotic material and culture of involved tissue and bone. Antimicrobial therapy alone is not effective for cure of infected and necrotic bone. • It is reasonable to consider deferring antimicrobial therapy initiation until bone and/or joint microbiological samples are obtained for clinically stable patients[1]. • Decisions regarding the delay of empirical therapy therefore balance potential harm due to the risk of progression of life-threatening infection (e.g., sepsis) or impending spinal cord compression against the potential benefit of microbiological data[1]. • For **IV to oral conversion**, see notes below.

CHAPTER 3.1	BONE AND JOINT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
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Chronic osteomyelitis Therapy should be based on bone biopsy culture **Most common:** *S. aureus* **Definition:** • Relapsing infection despite adequate duration antibiotics. • Chronic pain/ swelling/bone tenderness associated with tissue necrosis, increased drainage, or persistent sinus tracts. • Presence of bone destruction and presence sequestra on imaging.	Empirical antibiotics prior cultures are not recommended.		• **Duration:** • Minimum duration 6 weeks. • To extend if clinical improvement is not sufficient. • If osteomyelitis is fully resected (i.e., amputation), stop antibiotics when the operative wound is healing. • Recommended to send histopathological examination (HPE) of the clean margin. • Ideally therapy should be based on bone biopsy culture as swab culture is not reliable. Culture from bone biopsy is vital for recurrent disease. • Thorough surgical debridement of sequestra or foreign body. Can consider addition of local delivery of antibiotics based on available guidelines as an adjunct to systemic antibiotic (e.g., antibiotic impregnated collagen sheets or antibiotic laden cement)[1]. • Decision for other method of local delivery of antibiotics should be based on a multidisciplinary team (MDT) discussion with the infectious disease (ID) team.

Vertebral osteomyelitis With or without epidural extension *S. aureus*	Cloxacillin 2 g IV q4H	Cefazolin 2 g IV q6-8H	• Empirical therapy should be withheld unless the patient is septic or in patients with neurologic compromise. • Empiric gram-negative should be covered if patient had: • Recent spinal hardware inserted or surgery neurologic compromise. • Intraabdominal infections • Coexisting or synchronous genitourinary infection • HIV infection • Investigate for brucellosis or tuberculosis if clinically indicated. • Evaluation for concurrent infective endocarditis is warranted in patients with underlying valvular disease and/or new-onset heart failure in the setting of positive blood cultures for gram-positive organisms[2]. • To tailor antibiotics according to definitive cultures. • To evaluate patient if surgical therapy is necessary, such as[2]: • Spinal cord compression/instability • Persistence of epidural abscess and paravertebral abscess • Considering TB spine/MDR organisms • Progression, persistence, or recurrence of disease (as documented by persistently positive blood cultures or worsening pain) despite appropriate antimicrobial therapy. • In the absence of the above conditions warranting surgical intervention, patients with radiographic evidence of vertebral osteomyelitis should undergo a CT-guided needle biopsy of the affected bone and disc space as well as aspiration of abscess if present based on discretion of the surgeon. The specimens should be sent for bacterial (aerobic and anaerobic), fungal, and mycobacterial culture as well as histologic examination[2]. • **Duration:** • Minimum 6 weeks. • If undrained paravertebral abscess(es) and/or infection due to drug-resistant organisms: Minimum 8 weeks • If extensive bone destruction: Up to 12 weeks. • For **IV to oral conversion**, see notes below.

Infections and Likely Organisms	Suggested Treatment		Notes
	Preferred	Alternative	
Fracture fixation infection/Osteosynthesis-associated infection (OAI)			
Early or acute[3]	Ampicillin/Sulbactam 3 g IV q8H ± Vancomycin 15 mg/kg (max 1 g) IV q8-12H*		• Debridement should be done and tissue/bone samples sent for culture and HPE preferably before antibiotic therapy is initiated. • **Duration:** • **Acute infection (within 3 weeks after implant insertion):** ○ Antibiotics are required until bone union (minimum of 6 weeks) or bone heals sufficiently enough that the implants can be removed in cases of fracture[1]. • **Delayed infection (≥ 3 weeks after implant insertion):** ○ Antibiotics are required for as long as the implant is present and until fractures are united[4]. • Rifampicin should be added if there is staphylococcal infection after the wound is clean and dry (after 5-7 days). • In the presence of delayed infection, implant is recommended to be removed.
Delayed[3]	Ceftazidime 2 g IV q8H + Vancomycin 15 mg/kg (max 1 g) IV q8-12H* ± Metronidazole 500 mg IV q8H		• In the event of bone union is not achieved, it is still recommended to remove the implant and consider alternative fracture stabilisation as per surgeon practice. Continue the antibiotics for 1 week post removal. • In circumstances where there are recurrent infections and implant removal is not feasible, long term antimicrobial suppression may be considered[4]. Decision for long term suppression should be made based on an MDT discussion with the ID team. • *Consider Vancomycin in septic patients, known MRSA-carriers, multiple previous surgeries, suspected low-grade infection[3].

Infections and Likely Organisms	Suggested Treatment		Notes
	Preferred	Alternative	
Paronychia/Felon			
Bacterial	Cefalexin 500 mg PO q6H	Co-amoxiclav 625 mg PO q8H*	• In patients with risk factors for MRSA colonization, oral agents that cover MRSA are a reasonable choice: • Co-trimoxazole 960-1920 mg PO q12H • *For patients with a digit that has been exposed to oral flora (nail biting, finger sucking), antibiotic coverage should include *S. aureus, E. corrodens, Haemophilus influenzae*, and beta-lactamase-producing oral anaerobic bacteria. • For patients with deep seated infection and systemic symptoms for surgical debridement and consider IV antibiotics. • Duration: 5-10 days
Viral *Herpes simplex* (Whitlow)	Aciclovir 400 mg PO q8H		• Duration: 10 days
Fungal Dishwasher (prolonged water immersion) *Candida spp.*	Miconazole LA q6-8H	Itraconazole 200 mg PO q24H	• Duration: 7-14 days
Other infections			
Soft tissue trauma, crush injuries, stab wounds	Cefazolin 2 g IV q8H	Cefuroxime 1.5 g IV q8H	• Duration: 5 – 10 days. • Add Metronidazole 500 mg IV q8H ± Gentamicin 3 – 5 mg/kg IV q24H if contaminated wound, abdominal injury, or devitalized tissue. • Recommended to perform a thorough surgical debridement of soft tissue.

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Flexor tenosynovitis *S. aureus* *Streptococcus* *Gram-negative rods*	Cloxacillin 2 g IV q6H ± Gentamicin 3-5 mg/kg IV q24H	Cefazolin 1-2 g IV q8H	• Duration: 5 – 10 days (subject to clinical assessment).
Open fractures[5] Refers to preoperative or when patient presents before definitive fixation.	Cefazolin 2 g IV q8H + Metronidazole 500 mg IV q8H ± Gentamicin 3-5 mg/kg IV q24H*	Cefuroxime 1.5 g IV q8H + Metronidazole 500 mg IV q8H ± Gentamicin 3-5 mg/kg IV q24H* **OR** Co-amoxiclav 1.2 g IV q8H**	• Addition of Metronidazole ± Gentamicin depends on severity and exposure causing wound contamination. • ** Use Co-amoxiclav if there is a significant contamination. • No need to obtain cultures in the setting of superficial debridement of necrotic tissue. • Pre-debridement and post debridement cultures are not representative of actual infection. • If subsequent debridement is performed for suspected infection, bone and tissue culture is warranted. • Prompt administration of antibiotics (ideally within six hours) is warranted to reduce the risk of osteomyelitis[6]. • **Duration:** • Gustilo type I and II open fractures: discontinue at 24 hours after wound closure. • Gustilo type III open fractures: discontinue at 72 hours after wound debridement. • Duration of antibiotic may need to be extended if wound closure is delayed. Delay in wound closure increases the risk of MDRO, therefore all effort should be made to close the wounds as early as possible.

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Spinal implant infection Therapy should be based on culture results	Bone culture is warranted prior starting antibiotics. Treatment should be based on culture data obtained from wound drainage and/or operative debridement.		• **Early onset (< 90 days)[7]:** • Debridement and retain hardware (90% success): ○ 2 weeks of IV antibiotic, minimum duration of total antibiotic is 12 weeks (maximum duration is guided by clinical, biochemical, and radiological progress). • **Late onset (> 90 days)[7]:** • Debridement and remove hardware: ○ Minimum 2 weeks of IV antibiotic, minimum duration of total antibiotic is 12 weeks (maximum duration is guided by clinical, biochemical, and radiological progress). ○ If removal of the hardware following union is feasible, repeat cultures should be obtained at the time of hardware removal to guide subsequent therapy, which typically includes at least another 6 weeks of antimicrobial therapy[2]. • Debridement and hardware retained: ○ Minimum 2 weeks of IV antibiotic and total duration of antibiotic is until there is fusion (the hardware will be removed if there is clinical sign of ongoing infection). Maximum duration is guided by clinical, biochemical, and radiological progress. ○ If hardware removal is not feasible, consider antimicrobial suppression (tailored to culture and susceptibility data)[2].

References:

1. Use of Novel Strategies to Develop Guidelines for Management of Pyogenic Osteomyelitis in Adults. JAMA Network Open, 5(5), e2211321. https://doi.org/10.1001/jamanetworkopen.2022.11321
2. Pocket Guide to Diagnosis and Treatment of implant-associated infections after fracture fixation. Version 2. PRO-IMPLANT Foundation (N. Renz, A. Trampuz).
3. UpToDate: Nonvertebral osteomyelitis in adults: Treatment. Date accessed: 20 March 2024.
4. Standards for the Management of Open Fractures. Oxford University Press. https://doi.org/10.1093/med/9780198849360.001.0001
5. UpToDate: Osteomyelitis associated with open fractures in adults. Date accessed: 20 March 2024.
6. Part IV: Spine. Proceedings of the second international consensus meeting (ICM) on musculoskeletal infection. 2018.

Prosthetic Joint Infection

Bacterial Aetiology		
Early onset (<3months after surgery)	**Delayed onset (3-24 months after surgery)**	**Late onset (>24 months after surgery)**
S. aureus Aerobic Gram-negative bacilli Anaerobes Mixed infections	Coagulase-negative Staphylococcus Enterococcus Propionibacterium (Cutibacterium)	*S. aureus* Beta-haemolytic streptococci Aerobic Gram-negative bacilli

Important Points

1. Give antibiotics (including surgical prophylaxis) only after collecting specimens.
2. Treatment strategy and duration of treatment depend on organism and surgical strategy.
3. Decision to treat, total duration and choice of antibiotic after the second stage procedure would depend on the culture result of the second stage procedure. If there is no growth in the latest intraoperative culture or culture is negative, decision for treatment with antibiotics (choice and duration) should be made based on an MDT with the ID team.
4. Tailor treatment according to clinical, biochemical, and radiological improvement.
5. Rifampicin should only be added for Staphylococcal infection only if the old prosthesis is retained or new prosthesis is implanted. Add it to the intravenous treatment as soon as wounds are dry and drains have been removed (after 5-7 days).
6. Prior to adding Rifampicin, to check if *Staphylococcus* spp. is sensitive towards Rifampicin.

CHAPTER 3.1	BONE AND JOINT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Prosthetic joint infection			
Directed treatment			
Methicillin-sensitive *Staphylococcus* spp. (MSSA/MSSE)	Cloxacillin 2 g IV q4-6H + Rifampicin 600 mg PO q24H for 2-6 weeks **Followed by:** Rifampicin 600 mg PO q24H + Cefalexin 500 mg-1000 mg PO q6H **OR** Cloxacillin 1000 mg PO q6H	Cefazolin 2 g IV q8H **OR** Clindamycin 600 mg IV q8H + Rifampicin 600 mg PO q24H for 2-6 weeks **Followed by:** Rifampicin 600 mg PO q24H + Cefalexin 500-1000 mg PO q6H **OR** Cloxacillin 1000 mg PO q6H	• **Duration of antibiotics:** **a. Debridement and Implant Retention (DAIR):** 12 weeks. **b. One-Stage Exchange Strategy:** 12 weeks. **c. Two-Stage Exchange Strategy:** 12 weeks in between two-stage surgeries. • To discuss with primary team regarding the type (cement or implant) and duration of spacer used to decide on the need for Rifampicin. • If prolonged therapy is needed (beyond 12 weeks), stop Rifampicin and continue with the other oral agent being used as monotherapy. • For patients with staphylococcal PJI who cannot take Rifampicin because of drug resistance, allergy, toxicity, intolerance, or drug-drug interactions, antibiotic therapy should consist of pathogen-specific intravenous therapy for 4-6 weeks (before transitioning to oral antibiotic suppression, if warranted)[1].

Infections and Likely Organisms	Suggested Treatment		Notes
	Preferred	**Alternative**	
Methicillin-resistant *Staphylococcus* spp. (MRSA/MRSE)	Vancomycin 15 mg/kg (max 1 g) IV q8-12H + Rifampicin 600 mg PO q24H **Followed by:** Rifampicin 600 mg PO q24H + Sodium Fusidate 500 mg PO q8H		• **Duration of antibiotics:** **a. Debridement and Implant Retention (DAIR):** 12 weeks. **b. One-Stage Exchange Strategy:** 12 weeks. **c. Two-Stage Exchange Strategy:** 12 weeks in between two-stage surgeries. • To discuss with primary team regarding the type (cement or implant) and duration of spacer used to decide on the need for Rifampicin. • If prolonged therapy is needed (beyond 12 weeks), stop Rifampicin and continue with the other oral agent being used as monotherapy. • For patients with staphylococcal PJI who cannot take Rifampicin because of drug resistance, allergy, toxicity, intolerance, or drug-drug interactions, antibiotic therapy should consist of pathogen-specific intravenous therapy for 4-6 weeks (before transitioning to oral antibiotic suppression, if warranted)[1].
Beta-haemolytic *Streptococci*	Benzylpenicillin 20-24 MU IV daily divided q6H	Cefazolin 2 g IV q8H **OR** Clindamycin 600 mg IV q8H	• **Duration of antibiotics:** **a. Debridement and Implant Retention (DAIR):** 12 weeks. **b. One-Stage Exchange Strategy:** 12 weeks. **c. Two-Stage Exchange Strategy:** 12 weeks in between two-stage surgeries. • Decision for suppressive therapy should be made based on an MDT with the ID team.

Enterococcus		• **Duration of antibiotics:** **a. Debridement and Implant Retention (DAIR):** 12 weeks. **b. One-Stage Exchange Strategy:** 12 weeks. **c. Two-Stage Exchange Strategy:** 12 weeks in between two-stage surgeries. • There should be a high tendency for stopping Gentamicin in patients with deteriorating renal function or other signs of toxicity. • *Ceftriaxone should only be used for Gentamicin-resistant *Enterococcus faecalis* and in combination with another antibiotic. • Tailor treatment according to clinical, biochemical, and radiological improvement. • To send blood samples for TDM of Gentamicin at the third dose.
Penicillin-sensitive	Benzylpenicillin 20-24 MU IV daily divided q6H **OR** Ampicillin 2 g IV q4-6H ± Gentamicin 3-5 mg/kg IV q24H **OR** Ceftriaxone 2 g IV q24H* **Followed by:** Amoxicillin 1 g PO q8H[2]	
Penicillin-resistant	Vancomycin 15 mg/kg (max 1 g) IV q8-12H **Followed by:** Linezolid 600 mg PO q12H[2]	

Infections and Likely Organisms	Suggested Treatment		Notes
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Pseudomonas aeruginosa	Ceftazidime 2 g IV q8H **Followed by:** Ciprofloxacin 750 mg PO q12H[2]	Cefepime 2 g IV q8H **Followed by:** Ciprofloxacin 750 mg PO q12H[2]	• **Duration of antibiotics:** **a. Debridement and Implant Retention (DAIR):** 12 weeks. **b. One-Stage Exchange Strategy:** 12 weeks. **c. Two-Stage Exchange Strategy:** 12 weeks in between two-stage surgeries. • If the *Pseudomonas* is resistant to Ceftazidime, please refer to ID team. • *To check sensitivity of *Pseudomonas* towards Ciprofloxacin.
Cutibacterium acnes Shoulder infection is the most common presentation of PJI due to *Cutibacterium acnes* (formerly *Propionibacterium acnes*).	Benzylpenicillin 20 MU IV daily divided q4H **Followed by:** Phenoxymethylpenicillin 1 g PO q6-12H[3] **OR** Amoxicillin 1 g PO q8H[3]	Ceftriaxone 2 g IV q24H* **OR** Clindamycin** 600 mg IV q8H **OR** 450 mg PO q8H **Followed by:** Doxycycline 100 mg PO q12H[3] **OR** Cefalexin 500 mg -1 g PO q8H[3]	• **Duration of antibiotics:** **a. Debridement and Implant Retention (DAIR):** 12 weeks. **b. One-Stage Exchange Strategy:** 12 weeks. **c. Two-Stage Exchange Strategy:** 12 weeks in between two-stage surgeries. • *Consider Ceftriaxone in case of allergy to Penicillin or for OPAT. • **May consider Clindamycin if allergic to beta-lactam. Prolonged use of Clindamycin is associated with increased risk for *Clostridioides difficile* infection. To confirm susceptibility of *Cutibacterium acnes* to Clindamycin.

Culture negative[2] Please refer to ID team for any culture negative case.	Ampicillin/Sulbactam 3 g IV q8H ± Vancomycin 15 mg/kg (max 1 g) IV q8-12H **Followed by:** Rifampicin 450 mg PO q12H* + Levofloxacin 500 mg PO q12H	• **Duration of antibiotics:** **a. Debridement and Implant Retention (DAIR):** 12 weeks. **b. One-Stage Exchange Strategy:** 12 weeks. **c. Two-Stage Exchange Strategy:** 12 weeks in between two-stage surgeries. • *Rifampicin should only be administered after the prosthesis is reimplanted. Add to intravenous treatment when wounds are dry and drains removed. For patient >75 years: reduce the dose to 300 mg PO q12H[2].
Fungal	Decision for treatment with antibiotics (including choice and duration) should be made based on an MDT with the ID team.	

References:

1. UpToDate: Prosthetic joint infection: Treatment. Date accessed: 20 March 2024.
2. Pocket Guide to Diagnosis & Treatment for the Periprosthetic Joint Infection (PJI). Version 11. PRO-IMPLANT Foundation, Berlin, Germany (N. Renz, A. Trampuz).
3. Osmon, D. R., Berbari, E. F., Berendt, A. R., Lew, D., Zimmerli, W., Steckelberg, J. M., Rao, N., Hanssen, A., & Wilson, W. R. (2012). Diagnosis and Management of Prosthetic Joint Infection: Clinical Practice Guidelines by the Infectious Diseases Society of America. Clinical Infectious Diseases, 56(1), e1–e25. https://doi.org/10.1093/cid/cis803.

Antibiotic Cement Implantation

General Concept[1]

1. The higher the antibiotic concentration added, the less the mechanical features are expected to be. The antibiotic-to-cement ratio depends on whether it is used as spacer or for fixation.
2. Fixation: <5% antibiotics to dry weight of polymethylmethacrylate (PMMA).
3. Antibiotic spacers: 10-15% antibiotics to dry weight of PMMA.
4. Vancomycin is the preferred choice if organisms are not identified. High doses can be delivered through the cement and is bactericidal at those doses even for gram-negatives. Once an organism is identified, targeted local antibiotic therapy can be used.

Doses of antibiotics to be added into bone cement with their attributed weight proportion per 40 g of the bone cement (low dose/high dose)[1]

Antibiotics	Low dose (g/40 g bone cement)	High dose (g/40 g bone cement)
Amikacin	1	2
Amphotericin B	0.2	N/A
Cefotaxime	2	3
Ciprofloxacin	0.2	3
Clindamycin	1-2	4,8
Cefazolin	1-2	4-8
Cefuroxime	1.5	3
Colistin	0.24	N/A
Daptomycin	2	N/A
Erythromycin	0.5	1
Gentamicin	0.25-1	4.8

Doses of antibiotics to be added into bone cement with their attributed weight proportion per 40 g of the bone cement (low dose/high dose)[1]		
Antibiotics	**Low dose (g/40 g bone cement)**	**High dose (g/40 g bone cement)**
Linezolid	1.2	N/A
Meropenem	0.5	4
Piperacillin/Tazobactam	4	8
Vancomycin	1	3-9

References:

1. Shahpari O, Mousavian A, Elahpour N, Malahias MA, Ebrahimzadeh MH, Moradi A. The Use of Antibiotic Impregnated Cement Spacers in the Treatment of Infected Total Joint Replacement: Challenges and Achievements. Arch Bone Jt Surg. 2020 Jan;8(1):11-20. doi: 10.22038/abjs.2019.42018.2141.

Intravenous to Oral Conversion

General Concept

- Most infections require completion of treatment with parenteral therapy.
- However, completion of treatment with oral therapy following at least 2 weeks of parenteral therapy may be reasonable in the following circumstances:
  1. Clinically stable (haemodynamically and at the site of infection, e.g., no spinal instability)
  2. The infection is uncomplicated, and the patient has no significant comorbidities.
  3. A favourable clinical response to initial parenteral therapy is observed (CRP is reducing).
  4. A suitable oral drug with proven susceptibility to the causative organism is available.
  5. Compliance with oral therapy can be assured or carefully monitored.
  6. Adequate source control (i.e., not requiring further procedural drainage and without persistent bacteraemia).
  7. Are likely to absorb oral medications from a functioning gastrointestinal tract.
  8. Have no psychosocial reasons that preclude the safe use of oral therapy.

Oral antibiotic options for conversion from intravenous (IV) route			
1.	Ampicillin/Sulbactam 750 mg q12H	**7.**	Metronidazole 400 mg q8H
2.	Cefalexin 500 mg – 1 g q6H	**8.**	Rifampicin 300 mg q12H or 600 mg q24H*
3.	Co-amoxiclav 625 mg q8H or 1 gm q12H	**9.**	Sodium Fusidate 500 mg q8H*
4.	Co-trimoxazole (480 mg) 4 tablets q12H or 8–10mg/kg/day of TMP dose in divided dose	**10.**	Clindamycin 600 mg q8H
5.	Doxycycline 100 mg q12H	**11.**	Ciprofloxacin 750 mg q12H**
6.	Linezolid 600 mg q12H		

* Must always be used in combination.

** Fluoroquinolones are associated with collateral damage (collateral damage means ecological adverse effects of antibiotic therapy; like selection of drug-resistant organisms and the unwanted development of colonization or infection with multidrug-resistant organisms and C. difficile infection). Besides, Fluoroquinolones are also associated with a number of adverse effects like tendonitis, tendon rupture, prolonged QTc which in turns leading to arrhythmia, delirium, dysglycaemia (high or low sugar), peripheral neuropathy and associated with aortic aneurysm and rupture. All are in the black box FDA warning. Reserve for use in patients who have no alternative treatment options.